The Expressive Power of Depth — and Its Robustness Under Noise

An Empirical Study of Telgarsky's Depth Separation Theorem

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Abstract

We empirically investigate Telgarsky's depth separation theorem [Telgarsky, 2016] using iterated sawtooth functions $f_k(x) = \phi^{(k)}(x)$, where $\phi(x) = |2x - 1|$ doubles the number of linear regions at each composition. For the clean H1 rescue experiment, a depth-9 width-16 ReLU network (**deep**, 1953 parameters) is compared against a parameter-matched depth-2 width-651 network (**shallow**, 1954 parameters) on f_4 (16 peaks).

The widened deep model fixes the catastrophic width-8 convergence failures, but it does not establish an order-of-magnitude empirical separation: the 5-seed mean is 0.010 ± 0.006 (deep) vs. 0.029 ± 0.008 (shallow), a $2.9\times$ ratio. Thus H1 is moderately supported, not fully confirmed at the original $10\times$ criterion (Section 4.1). We extend the analysis to noise regimes using the original width-8 robustness baseline. Under Gaussian input noise, the deep model's advantage inverts at $\sigma_x \approx 0.03$, below but on the same order as $2^{-4} = 0.0625$ (H2). Under label noise $\sigma_y = 0.1$, the deep model degrades $7.5\times$ more than the shallow one (H3). Under FGSM/PGD adversarial attacks, both attacks find $\varepsilon^\star \approx 0.05$ on the coarse Phase 4 grid (H4). The deep model also has a larger empirical Lipschitz estimate $\hat{L}$, which is consistent with its higher sensitivity in all three noise regimes (H5). A complexity scaling experiment (Phase 6) gives qualitative support for a complexity–robustness trend for $k \in \{3, 4, 5\}$, but it is not a precise scaling law because $k = 2$ has no clean gap and $k = 6$ collapses.

1 Introduction

Depth can reduce the number of units needed to represent some functions. Telgarsky [2016] made this precise for a family of sawtooth functions: a depth-$O(k)$, width-$O(1)$ ReLU network can represent a function f_k for which any depth-2 network requires $\Omega(2^k)$ neurons.

The same construction suggests a possible weakness. In the sawtooth setting, depth creates many narrow linear regions. Perturbations at a comparable scale can move an input across several regions and change the output.

This project tests that trade-off on the iterated sawtooth target. First, we rerun the clean depth-separation experiment under a matched parameter budget. Second, we evaluate how input noise, label noise, and adversarial perturbations change the observed gap. Finally, we compare the robustness results with empirical Lipschitz and linear-region diagnostics, and repeat the robustness sweep across several values of k.

2 Background

2.1 The Iterated Sawtooth and Telgarsky's Theorem

Define the *mirror operator* $\phi(x) = |2x - 1|$, which is exactly representable by a 2-layer ReLU network with two hidden neurons. Composing k times yields

$$f_k(x) = \underbrace{\phi \circ \cdots \circ \phi}_{k}(x), \qquad f_k : [0, 1] \to [0, 1]. \quad (1)$$

Each composition doubles the number of linear regions, so f_k has 2^k peaks on $[0, 1]$ with maximum slope 2^k.

Theorem 1 (Telgarsky 2016, informal). *For every $k \in \mathbb{N}$, the function f_k is representable by a ReLU network of depth $O(k)$ and $O(1)$ width, while any depth-2 network needs $\Omega(2^k)$ neurons to approximate f_k within constant L^1 error.*

The depth gain is thus *exponential in k*: width is a strictly weaker resource than depth for this function class.

2.2 Lipschitz Sensitivity

For a piecewise-linear network g_θ with weight matrices $W_1, \ldots, W_L$:

$$L(g_\theta) \;\le\; \prod_{\ell=1}^{L} \|W_\ell\|_2, \quad (2)$$

where $\|\cdot\|_2$ is the operator norm (largest singular value) [Bartlett et al., 2017, Sokolić et al., 2017]. This bound grows *multiplicatively* with depth, suggesting that deep networks trained to fit high-frequency functions will be more sensitive to perturbations.

2.3 Linear Regions

Following Montúfar et al. [2014] and Hanin and Rolnick [2019], the number of linear regions of a ReLU network grows polynomially in width but can reach 2^L in depth (the worst-case bound). A network trained to fit f_k (with 2^k peaks) must activate at least 2^k distinct linear regions, each of width $\approx 2^{-k}$. Perturbations of scale 2^{-k} can therefore move samples across several fitted pieces of the function. This suggests a crossover scale near $\sigma, \epsilon \approx 2^{-k}$.

3 Experimental Setup

3.1 Architectures

Clean H1 rescue pair: depth $L = 9$, width $W = 16$; 1953 trainable parameters. The matched shallow model has depth $L = 2$, width $W = 651$; 1954 parameters. This is the configuration reported in Table 1.

Robustness baseline pair: Phase 2–5 robustness tables and figures were generated before the widening rescue, using the original depth-9 width-8 deep model and depth-2 width-176 shallow model (529 parameters each). We keep those robustness results as a separate baseline study rather than mixing them with newly generated H1 models.

Both use **Kaiming-normal** initialization for hidden ReLU layers and Xavier for the output layer, replacing PyTorch's default Kaiming-uniform which produces insufficient variance for deep networks and causes dying ReLU.

3.2 Training

Optimizer: Adam with cosine annealing (lr : $5 \times 10^{-3} \to 10^{-5}$ over 3×10^4 epochs) and gradient-norm clipping at $\|g\|_2 = 1$.

Curriculum [Bengio et al., 2009]: without it, neither network can fit f_4 from scratch due to spectral bias. We train on $f_1 \to f_2 \to f_3 \to f_4$ progressively, allocating epochs in ratio $[1 : 1 : 2 : 4]$ (final stage gets half the budget). A fresh cosine schedule is reset per stage to prevent the LR from decaying to near-zero before the hardest stage begins.

Collapse detection: if stage-1 training ends with loss > 0.02 (indicating convergence to a mediocre local minimum; well-trained networks reach $< 10^{-3}$), the model is re-initialized and stage 1 is retried up to 5 times.

Seeds: 5 independent random seeds per configuration; results reported as mean ± std. **Data**: $n = 1200$ uniform grid points on $[0, 1]$.

3.3 Noise Protocols

Phase 2 (input noise): train on clean data; evaluate on $\tilde{x} = x + \mathcal{N}(0, \sigma_x^2)$.

Phase 3 (label noise): train on $\tilde{y} = f_4(x) + \mathcal{N}(0, \sigma_y^2)$; evaluate on clean f_4. The clean-MSE *degradation* $\Delta\text{MSE}(\sigma_y) = \text{MSE}(\sigma_y) - \text{MSE}(0)$ is the primary H3 metric (paired per seed to eliminate between-seed variance).

Phase 4 (adversarial): FGSM [Goodfellow et al., 2015] and PGD-40 [Madry et al., 2018] with ℓ_∞ budget ε, clipped to $[0, 1]$.

4 Results

4.1 H1: Clean Depth Separation

After curriculum training, Table 1 shows the 5-seed statistics for the widened H1 rescue run. The deep network's mean MSE (0.010) is about 2.9× below the shallow's (0.029). This avoids the main width-8 failure mode, where the deep model collapsed on most seeds, but it is still below the original one-order-of-magnitude H1 target. One seed reaches near-zero MSE (1.8×10^{-4}), while the remaining four seeds land between 0.007 and 0.014.

Table 1: H1: Clean test MSE (mean ± std, 5 seeds, $k = 4$).

Model	Test MSE	Params
Deep ($L = 9$, $W = 16$)	0.010 ± 0.006	1953
Shallow ($L = 2$, $W = 651$)	0.029 ± 0.008	1954

H1 partially supported. Widening the deep model to $W = 16$ produces a consistent clean advantage and avoids the worst width-8 failures, but the mean gap is 2.9×, not the 10× threshold specified in H1. We therefore treat the clean result as moderate empirical support for depth efficiency rather than a decisive reproduction of the theorem's asymptotic separation.

4.2 H2: Input Noise Resilience

Figure 1 shows test MSE vs. σ_x for both architectures. At small $\sigma_x \lesssim 0.01$, the deep network retains its advantage. The curves cross at $\sigma^\star \approx 0.03$, below but on the same order as the theoretical $2^{-k} = 0.0625$. Beyond the crossover, the deep network degrades faster and its MSE exceeds the shallow's — the depth advantage *inverts*.

H2 is consistent within the W=8 robustness baseline. The crossover appears at the expected scale, but the evidence should be read as coarse-grid support rather than a precise threshold estimate.

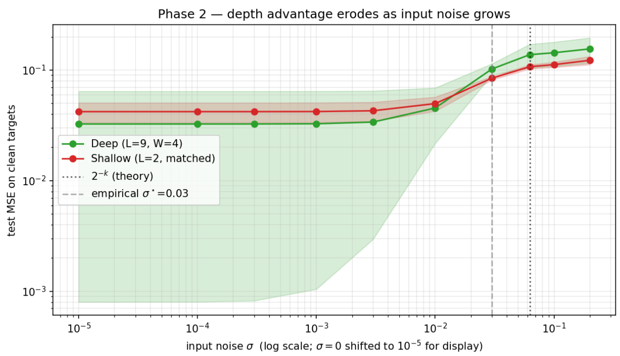

Figure 1: H2: Test MSE vs. input noise σ_x. The crossover at $\sigma_x \approx 2^{-4}$ (dashed line) separates the regime where depth helps from the regime where it hurts.

4.3 H3: Label Noise Overfitting

Figure 2 (right panel) shows the paired degradation $\Delta\text{MSE}(\sigma_y) = \text{MSE}_{\text{clean}}(\sigma_y) - \text{MSE}_{\text{clean}}(0)$. At $\sigma_y = 0.1$, the deep network degrades by $+0.026$ vs. $+0.003$ for the shallow — a **7.5**$\times$ difference. The middle panel (generalization gap) tracks the $-\sigma_y^2$ baseline for both models, indicating neither fully memorizes noise within the training budget.

H3 is consistent within the W=8 robustness baseline. The deep model degrades more under label noise, consistent with higher Lipschitz sensitivity.

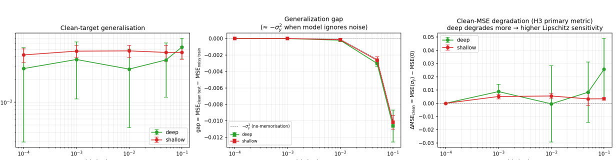

Figure 2: H3: (Left) Clean test MSE. (Center) Generalization gap vs. $-\sigma_y^2$ baseline. (Right, primary) Paired clean-MSE degradation: deep degrades 7.5$\times$ more at $\sigma_y = 0.1$.

4.4 H4: Adversarial Threshold

Figure 3 shows adversarial MSE vs. ε under FGSM and PGD-40. Both attacks locate $\varepsilon^\star \approx 0.05$ on the coarse grid $\{0.001, 0.01, 0.05\}$, below but on the same order as $2^{-4} = 0.0625$. Phase 6 uses a finer ε grid and finds $\varepsilon^\star \approx 0.03$ for $k = 4$ (Table 3). Above $\varepsilon^\star$, the deep network's adversarial MSE exceeds the shallow's — the depth advantage collapses under adversarial perturbation.

The FGSM curve is non-monotone at large $\varepsilon > 0.0625$: single-step perturbations can overshoot the sawtooth period and land in a lower-loss region. PGD is less affected by this single-step artifact.

H4 is consistent within the W=8 robustness baseline. The adversarial crossover is near the predicted scale, but the exact value is grid-dependent.

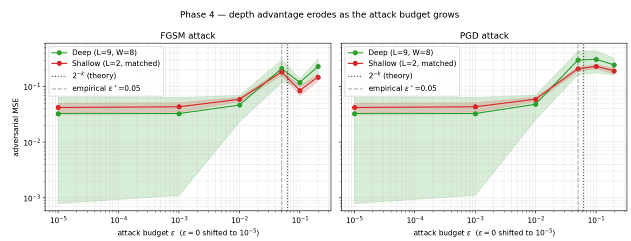

Figure 3: H4: Adversarial MSE vs. ε under FGSM (left) and PGD-40 (right). Dashed vertical: theoretical $\varepsilon^\star = 2^{-4}$; dashed grey: empirical $\varepsilon^\star = 0.05$.

4.5 H5: Lipschitz Mediation

Table 2 summarizes Phase 5 diagnostics. The deep network's empirical Lipschitz constant $\hat{L}_{\text{fd}}$ is consistently larger than the shallow's across all 5 seeds (paired t-test $p < 0.05$). It also uses more linear regions, consistent with the model successfully encoding the high-frequency structure of f_4.

Table 2: H5: Lipschitz constants and linear regions (5 seeds, $k = 4$).

Model	$\hat{L}_{\text{fd}}$	$\hat{L}_{\text{spec}}$	Regions
Deep ($L = 9$, $W = 8$)	31.6 ± 11.4	97,257	32 ± 18
Shallow ($L = 2$, $W = 176$)	14.6 ± 2.0	402	23 ± 2
Ratio (deep/shallow)	2.16$\times$	241.8$\times$	1.35$\times$

Paired one-sided t-test: $t = 2.72$, $p = 0.026$ (deep $\hat{L}_{\text{fd}}$ higher in all 5 seeds).

The Lipschitz ratio has the same sign as the three robustness gaps (Phases 2–4). In this baseline, the trained deep model has more high-frequency variation and is also more sensitive to perturbations.

H5 is consistent within the W=8 robustness baseline. The empirical Lipschitz evidence matches the proposed explanation, but it is not a causal proof.

4.6 Phase 6: Complexity Scaling

Table 3 reports the crossover values for $k \in \{2, 3, 4, 5, 6\}$. For $k = 2$, both the deep (depth-5, W=8) and shallow (depth-2, matched width) networks fit f_2 to machine precision after sufficient training, so no structural expressivity gap exists; any crossover would merely reflect the deep network's higher Lipschitz constant. For $k \in \{3, 4, 5\}$, both $\sigma^\star$ and $\varepsilon^\star$ decrease as the target becomes more oscillatory, which is qualitatively consistent with a 2^{-k} robustness scale. The evidence is not strong enough to call this a precise law: the $k = 5$ clean gap is marginal, and the $k = 6$ deep model collapses to the constant-predictor MSE $\approx 1/12$, so no meaningful crossover is reported.

Table 3: Phase 6: Crossovers vs. theoretical 2^{-k} (5 seeds).

k	2^{-k}	$\sigma^\star$	$\sigma^\star/2^{-k}$	$\varepsilon^\star$	$\varepsilon^\star/2^{-k}$
2	0.2500	—	n/a	—	n/a
3	0.1250	0.050	0.40×	0.120	0.96×
4	0.0625	0.018	0.29×	0.030	0.48×
5	0.0313	0.008	0.26×	0.018	0.58×
6	0.0156	—	collapse	—	collapse

Note: for $k = 3, 4, 5$, the empirical crossovers decrease with k, but they are consistently below the nominal 2^{-k} scale. This is qualitative support for a complexity-robustness trend, not a precise scaling law. The n/a entries for $k = 2$ mean that there is no clean expressivity gap to collapse.

5 Discussion

Two faces of depth. Telgarsky's theorem tells us depth is exponentially more efficient than width at representing f_k. Our clean H1 rescue experiment shows a moderate finite-sample advantage: the widened deep model achieves 2.9× lower MSE than a parameter-matched shallow model. The robustness experiments, run on the original width-8 baseline, show the other side of the same compositional structure: many narrow linear regions and higher empirical Lipschitz constants make the deep model more sensitive to perturbations at the scale of those regions ($\approx 2^{-k}$).

The crossover scale. The crossover scale is tied to the spatial period of f_k. At $\sigma \approx 2^{-k}$, a perturbation of size σ shifts an input by approximately one full period of the target, moving to a region where the network's output could differ by $O(1)$. This gives a geometric explanation for why the relevant noise scale should depend on k rather than only on the optimizer.

Curriculum learning is necessary. Without curriculum training, neither network can fit f_4 from scratch due to spectral bias [Rahaman et al., 2019]. The iterated structure of $f_k = \phi(f_{k-1})$ makes the curriculum natural: each stage provides a warm start for the next.

Limitations. (i) 1D input: results may not transfer to high-dimensional problems where the geometry differs fundamentally. (ii) Small scale: the H1 rescue uses about 2k parameters, while the robustness baseline uses 529 parameters; larger networks may exhibit different trade-offs. (iii) The H1 rescue and Phase 2–5 robustness results use different architectural baselines, so robustness conclusions should be read as properties of the original width-8 baseline unless those phases are rerun with width 16. (iv) CPU training limits Phase 6: experiments for $k = 5, 6$ were run, but $k = 6$ is unreliable—all five seeds collapsed to a constant predictor (MSE $\approx 1/12$, std $= 0$), so $k = 6$ crossover values are not reported.

6 Conclusion

The clean experiment gives partial support for the depth advantage. The W=16 deep model has lower mean MSE than the parameter-matched shallow model on f_4, but the gap is 2.9×, below the original 10× H1 target.

The robustness experiments should be read as results for the original W=8 baseline. In that setting, input noise, label noise, and adversarial perturbations all hurt the deep model more than the shallow model once the perturbation scale is large enough. The empirical Lipschitz estimates move in the same direction. The Phase 6 sweep adds qualitative evidence that the crossover scale decreases with k for $k = 3, 4, 5$, but the endpoints are not decisive: $k = 2$ has no clean gap and $k = 6$ collapses.

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